

Image Segmentation



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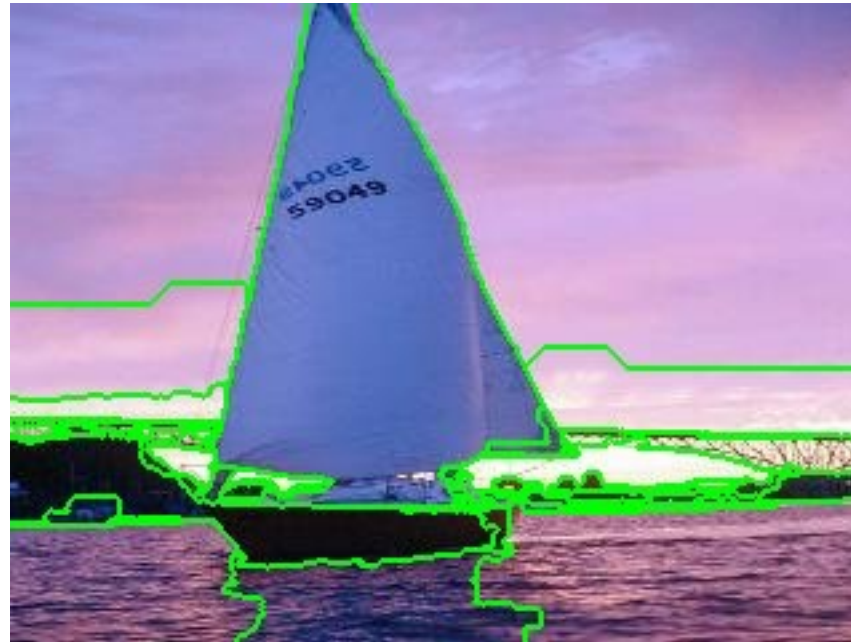
- What is image segmentation?
- Some research methods
- Case study: Remove Background

Image Segmentation

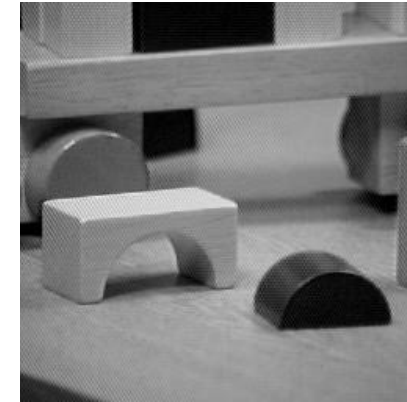
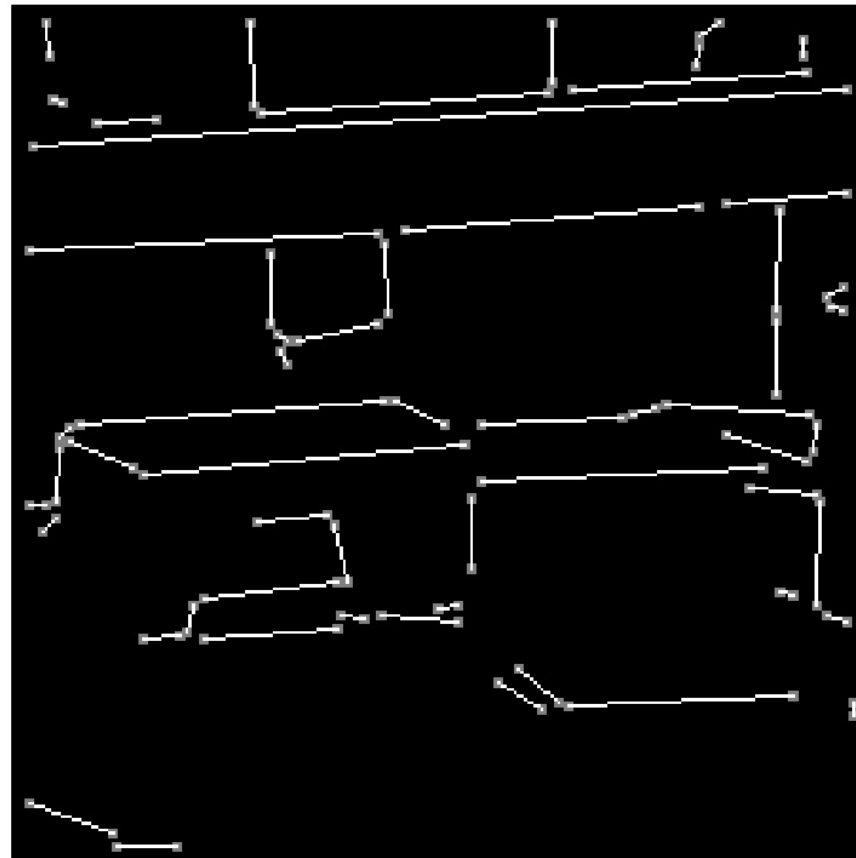
Image segmentation is the operation of partitioning an image into a collection of connected sets of pixels.

1. into **regions**, which usually cover the image
2. into **linear structures**, such as
 - line segments
 - curve segments
3. into **2D shapes**, such as
 - circles
 - ellipses
 - ribbons (long, symmetric regions)

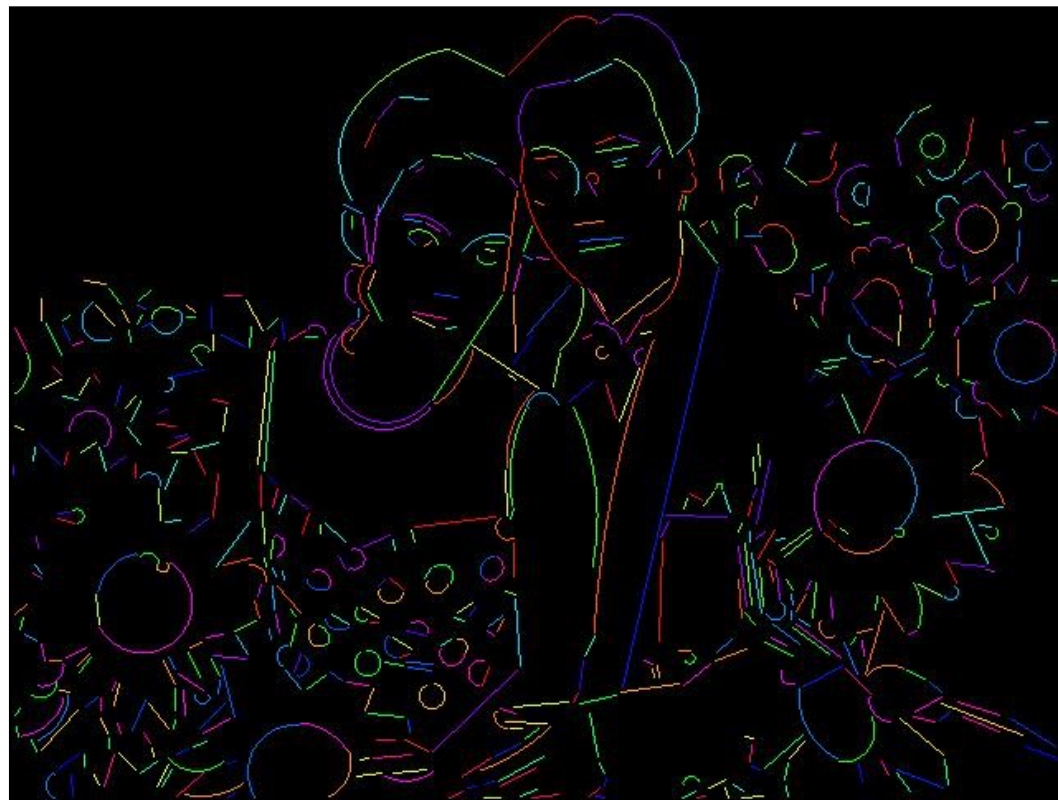
Example 1: Regions



Example 2: Straight Lines



Example 3: Lines and Circular Arcs



Region Segmentation: Segmentation Criteria

A segmentation is a partition of an image I into a set of regions S satisfying:

- | | |
|--|--|
| 1. $\cup S_i = S$ | Partition covers the whole image. |
| 2. $S_i \cap S_j = \phi, i \neq j$ | No regions intersect. |
| 3. $\forall S_i, P(S_i) = \text{true}$ | Homogeneity predicate is satisfied by each region. |
| 4. $P(S_i \cup S_j) = \text{false}, i \neq j, S_i \text{ adjacent } S_j$ | Union of adjacent regions does not satisfy it. |

From Pavlidis

So

So all we have to do is define and implement the similarity predicate.

But, what do we want to be similar in each region?

Is there any property that will cause the regions to be meaningful objects?

Main Methods of Region Segmentation

1. Region Growing
2. Clustering
3. Split and Merge

Case Study: Remove Background

Topic In Computer Vision

Problem Statement

Input: Image

Output: Image that only contain objects



Problem Statement



Research Methods

- Ostu Historgam
- Kmeans
- Graph Cuts

Questions and Answers